

## SpeedRadar2.py

```
1 import cv2
2 from tracker2 import *
3 import numpy as np
4 end = 0
5
6 #Creator Tracker Object
7 tracker = EuclideanDistTracker()
8
9 #cap = cv2.VideoCapture("Resources/traffic3.mp4")
10 cap = cv2.VideoCapture("traffic4.mp4")
11 f = 25
12 w = int(1000/(f-1))
13
14
15 #Object Detection
16 object_detector = cv2.createBackgroundSubtractorMOG2(history=None, varThreshold=None)
17 #100,5
18
19 #KERNELS
20 kernalOp = np.ones((3,3),np.uint8)
21 kernalOp2 = np.ones((5,5),np.uint8)
22 kernalCl = np.ones((11,11),np.uint8)
23 fgbg=cv2.createBackgroundSubtractorMOG2(detectShadows=True)
24 kernal_e = np.ones((5,5),np.uint8)
25
26 while True:
27     ret,frame = cap.read()
28     if not ret:
29         break
30     frame = cv2.resize(frame, None, fx=0.5, fy=0.5)
31     height,width,_ = frame.shape
32     #print(height,width)
33     #540,960
34
35
36     #Extract ROI
37     roi = frame[50:540,200:960]
38
39
40
41     #DIFFERENT MASKING METHOD
42     fgmask = fgbg.apply(roi)
43     ret, imBin = cv2.threshold(fgmask, 200, 255, cv2.THRESH_BINARY)
44     mask1 = cv2.morphologyEx(imBin, cv2.MORPH_OPEN, kernalOp)
45     mask2 = cv2.morphologyEx(mask1, cv2.MORPH_CLOSE, kernalCl)
46     e_img = cv2.erode(mask2, kernal_e)
47
48
49     contours,_ = cv2.findContours(e_img,cv2.RETR_TREE,cv2.CHAIN_APPROX_SIMPLE)
50     detections = []
51
52     for cnt in contours:
53         area = cv2.contourArea(cnt)
54         #THRESHOLD
55         if area > 1000:
56             x,y,w,h = cv2.boundingRect(cnt)
57             cv2.rectangle(roi,(x,y),(x+w,y+h),(0,255,0),3)
58             detections.append([x,y,w,h])
59
60     #Object Tracking
61     boxes_ids = tracker.update(detections)
```

```

62     for box_id in boxes_ids:
63         x,y,w,h,id = box_id
64
65
66         if(tracker.getsp(id)<tracker.limit()):
67             cv2.putText(roi,str(id)+" "+str(tracker.getsp(id)),(x,y-15), cv2.FONT_HERSHEY_PLAIN,
68 1,(255,255,0),2)
69             cv2.rectangle(roi, (x, y), (x + w, y + h), (0, 255, 0), 3)
70         else:
71             cv2.putText(roi,str(id)+ " "+str(tracker.getsp(id)),(x, y-15),cv2.FONT_HERSHEY_PLAIN
72 , 1,(0, 0, 255),2)
73             cv2.rectangle(roi, (x, y), (x + w, y + h), (0, 165, 255), 3)
74
75             s = tracker.getsp(id)
76             if (tracker.f[id] == 1 and s != 0):
77                 tracker.capture(roi, x, y, h, w, s, id)
78
79 # DRAW LINES
80
81 cv2.line(roi, (0, 410), (960, 410), (0, 0, 255), 2)
82 cv2.line(roi, (0, 430), (960, 430), (0, 0, 255), 2)
83
84 cv2.line(roi, (0, 235), (960, 235), (0, 0, 255), 2)
85 cv2.line(roi, (0, 255), (960, 255), (0, 0, 255), 2)
86
87 #DISPLAY
88 #cv2.imshow("Mask",mask2)
89 #cv2.imshow("Erode", e_img)
90 cv2.imshow("ROI", roi)
91
92 key = cv2.waitKey(w-10)
93 if key==27:
94     tracker.end()
95     end=1
96     break
97
98 if(end!=1):
99     tracker.end()
100
101 cap.release()
102 cv2.destroyAllWindows()

```